Q1

1

1. While loop runs n times
2. Insert shifts n elements
3. Thus time complexity is $O(n^2)$

2

1. While loop runs n times
2. Append takes constant time
3. Thus time complexity is O(n)

Q3

1. Makes an array of n items
2. First loop runs n times
3. Second loop runs n times
4. The loops are not nested
5. Thus time is O(n)

Q4

a)

1. While loop runs n times
2. Remove loops over n items
3. Thus time is $O(n^2)$

c)

1. First loop runs n times
2. Second loop runs n times
3. The loops are not nested
4. Thus time is O(n)